Statistics 280 Winter 2026 Midterm 2

Name (Please print legibly)	UMID Number

- Please answer all questions in the space provided and indicate where in the exam you have written additional work.
- **Solutions without justification will be marked as wrong for all questions except for questions asking for R code. Show all work. A calculator command does not count as justification**.
- Express answers as fractions or as decimals rounded to 2 decimal places.
- The last page is a table of values from the normal distribution.

Academic Integrity

Please read and sign the following statement regarding academic integrity to indicate that you have read, understand, and comply with the statement.

I have never given nor received unauthorized aid on this examination, nor have I concealed any violations of the academic integrity policy as stated by the College of Literature, Science, and the Arts. I have not done anything that may be understood as any action or attempted action that may result in creating an unfair academic advantage for oneself or an unfair academic advantage or disadvantage for any other member of members of the academic community. I understand that if I as suspected of academic misconduct, the incident may be reported to the Office of the Assistant Dean.

Signature: _____

1.(30 points) The popular television reality TV show **The Traitors** starts with 19 participants, with 3 randomly chosen participants designated as *traitors* and the remaining 16 participants designated as *faithfuls*. The traitors "murder" a faithful each evening starting Day 0, while all participants "banish" a participant during the data starting on Day 1. See the visual below. If there is only one traitor left at any point during the 3 following days, then the show randomly converts a remaining faithful into a traitor.

START	Evening 0	Day 1	Evening 1	Day 2	Evening 2	Day 3	Evening 3
19 participants 3 traitors 16 faithfuls	**Murder**	**Banish**	**Murder**	**Banish**	**Murder**	**Banish**	**Murder**

a) What is the chance the first participant banished from the game will be a traitor, assuming that the participants randomly choose whom to banish? Note that there are 18 participants remaining before the first banishment.

b) What is the chance that the first 6 participants removed from the game are all faithfuls? Assume that all removals from the show are at random. Note that the first, third, and fifth participants will be "murdered," and hence must be faithfuls.

c) What is the chance that at least one of the first 6 participants removed will be a traitor, assuming that all removals are at random?

d) What is the chance that exactly 2 of the first 6 participants removed will be traitors, assuming that all removals are at random?

2. (30 points) Suppose that a company with 100 employees has 50 employees who are women, 40 employees who are under 30 years of age, and 20 employees who are of Asian descent.

a) What is the maximum number of employees who are women, under 30 years of age, and of Asian descent? Draw a Venn diagram showing the maximum overlap of these 3 variables. The Venn diagram should label the number of employees in each part of the diagram.

b) Is it possible that no employees at the company are women, under 30 years of age, and of Asian descent? If so, then draw a Venn diagram with no overlap for these three groups. The Venn diagram should label the number of employees in each part of the diagram.

c) Suppose that there are 27 employees who are women and under 30 years of age. How many employees are women or under 30 years of age?

d) Is the number of women who are under 30 years of age in part c) more than, less than, or equal to the number of women you would expect to be under 30 years of age if being a woman and being under 30 years of age were independent? Explain.

3. (40 points) Suppose that 40% of non-graduating students at a university are first year students, 35% are second year students, and 25% third year students. The table below relates study habits to year.

	First year students	Second year students	Third year students
Studies fewer than 10 hours/week	50%	30%	20%
Studies at least 10 hours and fewer than 20 hours/week	35%	45%	40%
Studies at least 20 hours/week	15%	25%	40%

a) What percentage of non-graduating students study at least 20 hours/week?

b-d) What percentage of non-graduating students who study at least 20 hours/week are: b) first year, c) second year, and d) third year students?

4. (30 points) Suppose that a box contains 20 batteries, with 12 new batteries and 8 used batteries.

a) A technician randomly selects 5 batteries from the box all at once and tests them. What is the chance that exactly 2 of the batteries are new?

b) Write the R command to calculate the chance in part a). Include all inputs in your R command. _____.

c) A second technician selects 5 batteries randomly from the box one at a time, putting the battery back in the box after testing. What is the chance that more than 2 of the batteries are new?

d) Write the R command to calculate the chance in part c). Include all inputs.

_____.

5. (40 points) Suppose that you toss a fair coin 3 times and record the number of heads *n*. You then draw *n+1* tickets without replacement from a box with 4 tickets numbered 1, 1, 2, and 4 and take the maximum *M* of the numbers on those tickets.

a) Give a table with the possible values for *M* along with the probabilities for those values.

b) What is the expected value of *M*?

c) Write an R function called *simM()* to simulate the process of choosing *M* once. Note that the R function *max(v)* finds the maximum value in a vector *v*.

d) What is the following code estimating?

```
ms = c()
for (i in 1:10000) {
        ms = c(ms,simM())
}
print(sd(ms)*9999/10000)
```

6. (30 points) Your friend claims that half of the balls in a box are red and the other half blue. You pick 150 balls from the box with replacement and draw 100 red balls.

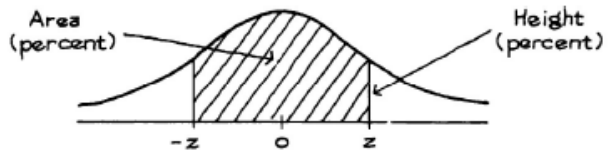
a) Write an R command to find the exact probability that you would draw between 70 and 100 red balls, assuming that your friend is telling the truth.

b) Estimate the chance that you would draw between 70 and 100 red balls assuming that your friend is telling the truth and that the number of red balls is approximately normally distributed. Remember to use the continuity correction.

c) Estimate the chance that you would draw 100 or more red balls assuming that your friend is telling the truth and that the number of red balls is approximately normally distributed. Remember to use the continuity correction.

d) Do you believe that your friend is telling you the truth? Explain.

Tables



A NORMAL TABLE

z	*Height*	*Area*	*z*	*Height*	*Area*	*z*	*Height*	*Area*
0.00	39.89	0	1.50	12.95	86.64	3.00	0.443	99.730
0.05	39.84	3.99	1.55	12.00	87.89	3.05	0.381	99.771
0.10	39.69	7.97	1.60	11.09	89.04	3.10	0.327	99.806
0.15	39.45	11.92	1.65	10.23	90.11	3.15	0.279	99.837
0.20	39.10	15.85	1.70	9.40	91.09	3.20	0.238	99.863
0.25	38.67	19.74	1.75	8.63	91.99	3.25	0.203	99.885
0.30	38.14	23.58	1.80	7.90	92.81	3.30	0.172	99.903
0.35	37.52	27.37	1.85	7.21	93.57	3.35	0.146	99.919
0.40	36.83	31.08	1.90	6.56	94.26	3.40	0.123	99.933
0.45	36.05	34.73	1.95	5.96	94.88	3.45	0.104	99.944
0.50	35.21	38.29	2.00	5.40	95.45	3.50	0.087	99.953
0.55	34.29	41.77	2.05	4.88	95.96	3.55	0.073	99.961
0.60	33.32	45.15	2.10	4.40	96.43	3.60	0.061	99.968
0.65	32.30	48.43	2.15	3.96	96.84	3.65	0.051	99.974
0.70	31.23	51.61	2.20	3.55	97.22	3.70	0.042	99.978
0.75	30.11	54.67	2.25	3.17	97.56	3.75	0.035	99.982
0.80	28.97	57.63	2.30	2.83	97.86	3.80	0.029	99.986
0.85	27.80	60.47	2.35	2.52	98.12	3.85	0.024	99.988
0.90	26.61	63.19	2.40	2.24	98.36	3.90	0.020	99.990
0.95	25.41	65.79	2.45	1.98	98.57	3.95	0.016	99.992
1.00	24.20	68.27	2.50	1.75	98.76	4.00	0.013	99.9937
1.05	22.99	70.63	2.55	1.54	98.92	4.05	0.011	99.9949
1.10	21.79	72.87	2.60	1.36	99.07	4.10	0.009	99.9959
1.15	20.59	74.99	2.65	1.19	99.20	4.15	0.007	99.9967
1.20	19.42	76.99	2.70	1.04	99.31	4.20	0.006	99.9973
1.25	18.26	78.87	2.75	0.91	99.40	4.25	0.005	99.9979
1.30	17.14	80.64	2.80	0.79	99.49	4.30	0.004	99.9983
1.35	16.04	82.30	2.85	9.69	99.56	4.35	0.003	99.9986
1.40	14.97	83.85	2.90	0.60	99.63	4.40	0.002	99.9989
1.45	13.94	85.29	2.95	0.51	99.68	4.45	0.002	99.9991